

POLARIN

POLAR
RESEARCH
INFRASTRUCTURE
NETWORK

Technical documentation
Architecture Design Document

Versions

Version	Date	Comment	Responsible
0.1	2024-??-??	???.	Øystein Godøy

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1. Introduction

1.1. Background

POLARIN is an international network of polar research infrastructures and their services, aiming at addressing the scientific challenges of the polar regions. The network includes a wide array of complementary and interdisciplinary top level research infrastructures: Arctic and Antarctic research stations, research vessels and icebreakers operating at both poles, observatories, data infrastructures and ice and sediment core repositories.

POLARIN will provide integrated, challenge-driven, and combined access to these infrastructures to facilitate interdisciplinary research on complex processes. POLARIN has the following six specific objectives, coherent with the work packages, each with measurable outcomes:

1. Enable science for understanding and predicting key processes in polar regions.
2. Provide efficient challenge-driven transnational access (TA) to top level RIs in the polar regions.
3. Improve data services and provide customised data products.
4. Provide virtual access (VA) to data and data services.
5. Provide training for infrastructure users.
6. Advertise RI services and engage RI users.

1.2. Scope of document

This document describes the architecture of POLARIN Virtual Access. It is not an official deliverable of the project, but aims to establish a common foundation for the work in work packages 4 and 5.

NOTE

This document is not a formalised Architecture Document following a formalised approach like RM-ODP (Reference Model for Open Distributed Processing), but rather a lightweight architecture approach.

1.3. Intended audience

Data and system managers within POLARIN, in particular participants in POLARIN Work Packages 4 and 5, "Improvement of data services and customised data products" and "Provision of virtual access" respectively.

1.4. Applicable documents

- RD-1** Description of work as represented by the proposal.
 - RD-2** Use Case Document (in draft version)
-

RD-3 Concept document (not available yet)

2. Overall concept

An outline of the division of tasks between work package 4 and 5 is provided in [Figure 1](#). This diagram indicates that development of the human front end to the Polarin data catalogue is done within WP 4, while virtual access as such is provided in WP 5. The latter is done through compilation of the Polarin data catalogue (WP 5). This catalogue is utilised by the human front end developed in WP 4. The requirements that the human interface has to fulfil is outlined in [RD-2](#).

NOTE

The front end can add additional sources to the human front end, but will have to check for uniqueness using the identifiers of the datasets found in the Polarin data catalogue.

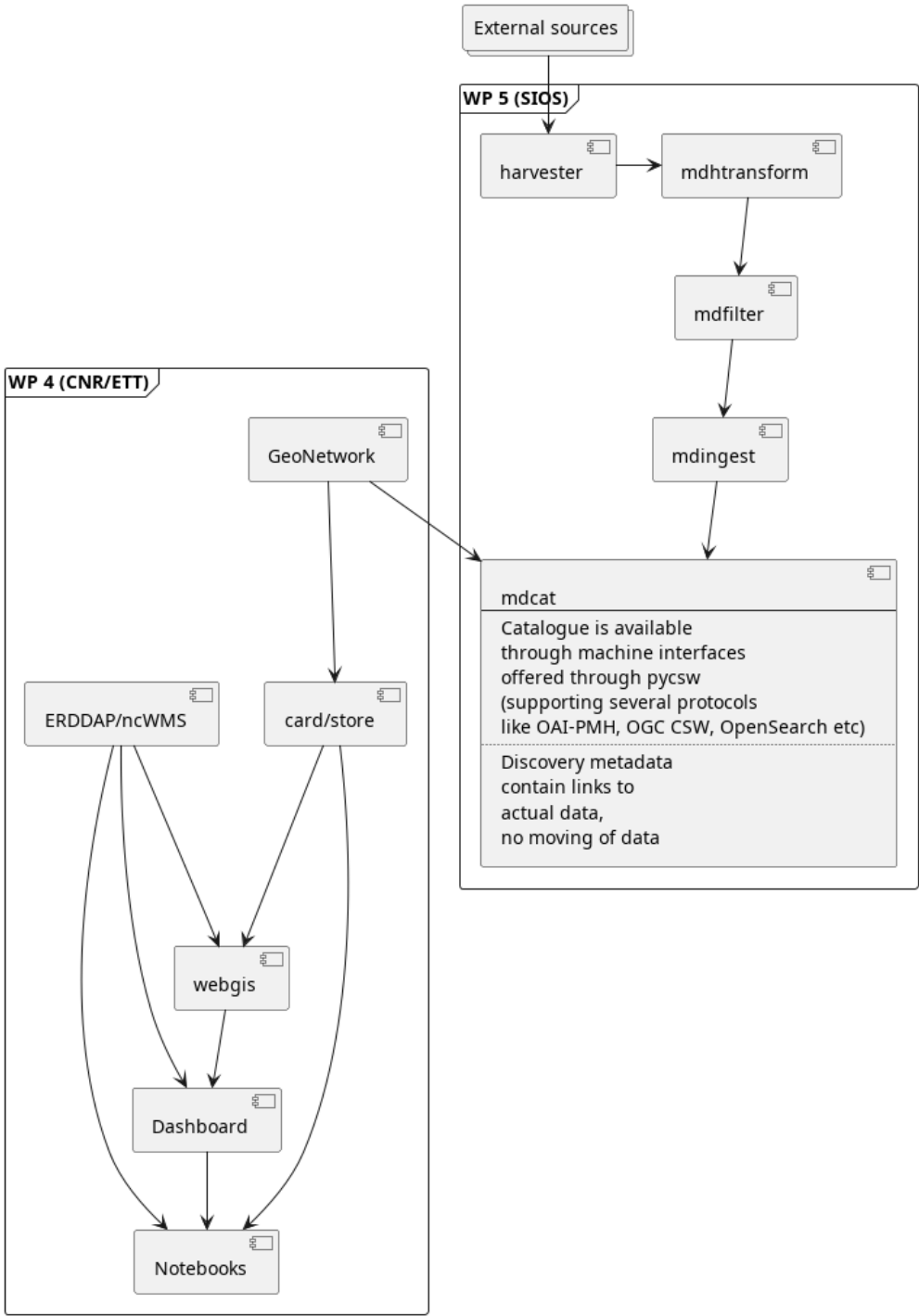


Figure 1. Architecture diagram

The datasets relevant for Polarin may be hosted in a number of data centres. Some of these are members of Polarin, while others are not but have been used by research infrastructures/stations for a long period. All datasets have to be properly tagged with Polarin as a project name. This information is used to filter out the relevant datasets for Polarin.

NOTE

A minimum discovery metadata profile has been identified and is further described below.

3. Detailed architecture

3.1. Architecture overview

NOTE

Should use [Figure 1](#) here and create a new above.

3.2. Data catalogue

FIXME

3.3. Human frontend

FIXME

4. Documentation standards

4.1. Minimum discovery metadata profile

A list of generic elements is provided in [Table 1](#), together with a high level explanation. In [Table 2](#) a suggested mapping between the metadata elements and commonly used metadata standards is provided.

NOTE

It is expected that a number of metadata standards has to be supported, including various profiles of ISO-19115, schema.org (preferably ESIIP’s science on schema.org), GCMD DIF, STAC etc.

Table 1. A minimum discovery metadata profile has been defined.

Element	Purpose
Metadata identifier	A unique identifier for the dataset. This is used to avoid duplicate records in aggregator catalogues. Utilisation of UUID with a namespace prefix is recommended.
Last update of metadata	An ISO8601 datetime for the last update of the metadata record.
Title	To provide a brief explanatory title for the dataset
Abstract	A short summary of the dataset, its purpose and how it was generated.
Temporal Extent	The temporal spanning of the dataset. This can be a multi temporal dataset.
Geographical Extent	The geographical location of the dataset. This can be a point, a bounding box, trajectory or a polygon.
Keywords	Keywords describing the dataset. Ideally this comes from controlled vocabularies and describes the variables of the dataset.
Personnel	Identification of all people that have contributed to the dataset. This requires full name, email, affiliation and a role description in the dataset. The latter has to come from a controlled vocabulary.
Publisher	This is identification of the data centre publishing the data. It contains a long and short name for the data centre and URL to the landing page.

Element	Purpose
Use constraint	This is a license for the data. This is a URL to the license text and an identifier. Utilisation of SPDX is recommended.
Data Access	This provides direct access to the dataset for download etc. It is not a landing page, but a direct link to the data and indication using a controlled vocabulary of the access mechanism (ranging from direct download to OPeNDAP and OGC WMS).
Project	A list of projects that has contributed to the creation of the dataset. Polarin has to be one of the projects.

Table 2. A suggested mapping between the elements of the minimum discovery metadata profile and some commonly used metadata standards.

Metadata field (Table 1)	ISO-19115 (ISO-19139, ISO-19115)	DIF 10.2 (DIF-10.2)	Schema.org (Schema.org)	DCAT (DCAT-2.0)	ACDD (ACDD)
Metadata identifier	gmd:fileIdentifier	Entry_ID	identifier	dct:identifier	id
Last update of metadata	gmd:dateStamp	Metadata_Dates/Metadata_Last_Revision	dateModified	dct:modified	date_metadata_modified
Title	gmd:identificationInfo/gmd:MD_DataIdentification/gmd:citation/gmd:CI_Citation/gmd:title	Entry_Title	name	dct:title	title
Abstract	gmd:identificationInfo/gmd:MD_DataIdentification/gmd:abstract	Summary/Abstract	description	dct:description	summary
Temporal Extent	gmd:identificationInfo/gmd:MD_DataIdentification/gmd:extent/gmd:EX_Extent/gmd:temporalElement/gmd:EX_TemporalExtent/gmd:extent/gml:TimePeriod/gml:beginPosition (gml:endPosition)	Temporal_Coverage/Range_DateTime/Beginning_Date_Time (Ending_Date_Time)	temporalCoverage	dct:temporal	time_coverage_start and time_coverage_end

Metadata field (Table 1)	ISO-19115 (ISO-19139, ISO-19115)	DIF 10.2 (DIF-10.2)	Schema.org (Schema.org)	DCAT (DCAT-2.0)	ACDD (ACDD)
Geographical Extent	<code>gmd:identificationInfo/gmd:MD_DataIdentification/gmd:extent/gmd:EX_Extent/gmd:geographicElement/gmd:EX_GeographicBoundingBox/gmd:westBoundLongitude (gmd:eastBoundLongitude, gmd:southBoundLatitude, gmd:northBoundLatitude)</code>	<code>Spatial_Coverage/Geometry/Bounding_Rectangle/Westernmost_Longitude (Northernmost_Latitude, Southernmost_Latitude, Easternmost_Longitude)</code>	<code>spatialCoverage</code>	<code>dct:spatial</code>	<code>geospatial_lat_min,geospatial_lat_max, geospatial_lon_min,geospatial_lon_max</code>
Keywords	<code>gmd:identificationInfo/gmd:MD_DataIdentification/gmd:descriptiveKeywords/gmd:MD_Keywords/gmd:keyword (and gmd:identificationInfo/gmd:MD_DataIdentification/gmd:descriptiveKeywords/gmd:MD_Keywords/gmd:thesaurusName/gmd:CI_Citation/gmd:title)</code>	<code>Science_Keywords for GCMD Science Keywords (or Ancillary_Key word for free text)</code>	<code>keywords</code>	<code>dcat:theme (or dcat:keyword for free text)</code>	<code>keywords, keywords_vocabulary</code>
Personnel	<code>gmd:identificationInfo/gmd:MD_DataIdentification/gmd:pointOfContact/gmd:CI_ResponsibleParty/gmd:individualName (gmd:organisationName, gmd:contactInfo and gmd:role)</code>	<code>Personnel/Role and Personnel/Contact_Person/First_Name (Last_Name, Email)</code>	<code>creator, contributor</code>	<code>dcat:contactPoint, dct:creator</code>	<code>creator_email, creator_institution, creator_name , creator_type, creator_url, contributor_name, contributor_role</code>
Publisher	<code>gmd:distributionInfo/gmd:MD_Distribution/gmd:distributor/gmd:MD_Distributor/gmd:distributorContact</code>	<code>Organization/Organization_Name/Short_Name (Long_Name) and Organization/Organization_URL</code>	<code>publisher</code>	<code>dct:publisher</code>	<code>publisher_name, publisher_email, publisher_institution, publisher_type, publisher_url</code>

Metadata field (Table 1)	ISO-19115 (ISO-19139, ISO-19115)	DIF 10.2 (DIF-10.2)	Schema.org (Schema.org)	DCAT (DCAT-2.0)	ACDD (ACDD)
Use constraint	<code>gmd:identificationInfo/gmd:MD_DataIdentification/gmd:resourceConstraints/gmd:MD_LegalConstraints/gmd:useConstraints</code> (type=otherRestrictions) and <code>gmd:identificationInfo/gmd:MD_DataIdentification/gmd:resourceConstraints/gmd:MD_LegalConstraints/gmd:otherConstraints</code>	Use_Constraints	license	dct:license	license
Data Access	<code>gmd:distributionInfo/gmd:MD_Distribution/gmd:transferOptions/gmd:MD_DigitalTransferOptions/gmd:online/gmd:CI_OnlineResource/gmd:linkage</code> (and <code>gmd:protocol</code>)	<code>Related_URL/URL_Content_Type/Type</code> and <code>Related_URL/URL</code>	Distribution and Distribution/contentUrl	<code>dcat:distribution</code> (<code>dcat:downloadURL</code> and <code>dcat:accessService</code>)	TBD
Project	can be provided through keywords using "project" as <code>gmd:MD_KeywordTypeCode</code>	Project/Short_Name (and Long_Name)	funding	TBD	project

4.2. Data documentation standards

Polarin is recommending usage of Darwin Core Archives and CF-NetCDF for documentation of datasets.

NOTE

Further details to be clarified.

FIXME

Darwin Core Archive

Is suitable for...

CF-NetCDF

Is suitable for...

NOTE

Data that doesn't fit into the classes above should be accompanied by a detailed documentation in PDF format.

4.3. Exchange of discovery metadata

FIXME